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Robotics MS, Computer Science concentration

Exercise 1: Multi-Armed Bandits

pics of questions 1, 2, 3 written by hand, showing work

HW1 RL

1) k-armed bandit, $k=4$ actions, bandit algorithm ϵ -greedy, sample-average action-value estimates and initial estimates of $Q_i(a)=0$, for all a .

Suppose initial sequence of actions and rewards

$$A_1=1, R_1=-1, A_2=2, R_2=1, A_3=2, R_3=-2, A_4=2, R_4=2, A_5=3, R_5=0$$

ϵ may have occurred, choosing action at random. When did it definitely occur?

When could it possibly occur?

$$Q_i(a) = \frac{\sum_{j=1}^{t-1} R_j \cdot \mathbb{1}_{A_j=a}}{\sum_{j=1}^{t-1} \mathbb{1}_{A_j=a}}$$

	A_1	A_2	A_3	A_4	A_5
1	1	1	0	0	0
2	0	1	1	1	0
3	0	0	0	0	1
4	0	0	0	0	0

first action must be random all Q_0 are equal

$$Q_i(a) = \sum_{j=1}^{t-1} R_j \cdot \mathbb{1}_{A_j=a} / \sum_{j=1}^{t-1} \mathbb{1}_{A_j=a}$$

$$Q_2(1) = \frac{(-1) \cdot 1}{1} = -1$$

$Q(1)$	0	-1	-1	-1	-1
$Q(2)$	0	0	1	-0.5	0
$Q(3)$	0	0	0	0	0
$Q(4)$	0	0	0	0	0

$$Q_n = \frac{R_1 + \dots + R_{n-1}}{n-1} \quad Q_{n+1} = Q_n + \frac{1}{n} [R_n - Q_n]$$

could have possibly taken random action for A_5 , technically any of them

on first step, always picks randomly

$$A=1 \quad n=1 \quad Q_2 = Q_1 + \frac{1}{2} [(R_1 - 1) - Q_1]$$

$$Q_2 = 0 + [-1 - 0] = Q_2(A=1) = -1$$

$$A=2 \quad n=1 \quad Q_3 = Q_2 + \frac{1}{1} [R_2 - Q_2]$$

$$Q_3 = -1 + [1 - (-1)] \quad Q_3 = 1$$

Second step, picks randomly if non tried actions

$$A=2 \quad n=2 \quad Q_4 = 1 + \frac{1}{2} [-2 - 1] = 1 - \frac{3}{2} = -\frac{1}{2} = -0.5$$

$$A=2 \quad n=3 \quad Q_5 = -0.5 + \frac{1}{3} [2 - (-0.5)] = -0.5 + \frac{1}{3} [\frac{5}{2} - \frac{1}{2}] = -0.5 + \frac{1}{3} [2] = -0.5 + 0.66 = 0.16$$

$$A=3 \quad n=1 \quad Q_6 = 0 + [0 - 0] \Rightarrow Q_6 = 0$$

2) Step size parameters, α_n , are not constant, then the estimate Q_n is a weighted average

$$Q_{n+1} = Q_n + \alpha [R_n - Q_n] \Rightarrow Q_{n+1} = \alpha R_n + (1-\alpha) Q_n$$

$$= \alpha R_n + (1-\alpha) [\alpha R_{n-1} + (1-\alpha) Q_{n-1}] \Rightarrow \alpha R_n + (1-\alpha) \alpha R_{n-1} + (1-\alpha)^2 Q_{n-1}$$

$$= (1-\alpha)^n Q_1 + \sum_{i=1}^n \alpha (1-\alpha)^{n-i} R_i \quad \text{different from this}$$

$$Q_{n+1} = Q_n + \alpha_n [R_n - Q_n] \Rightarrow Q_{n+1} = Q_n + \alpha_n R_n - \alpha_n Q_n \Rightarrow Q_{n+1} = \alpha_n R_n + (1-\alpha_n) Q_n$$

$$\Rightarrow Q_{n+1} = \alpha_n R_n + (1-\alpha_n) [\alpha_{n-1} R_{n-1} + (1-\alpha_{n-1}) Q_{n-1}]$$

$$Q_{n+1} = \alpha_n R_n + (1-\alpha_n) \alpha_{n-1} R_{n-1} + (1-\alpha_n)(1-\alpha_{n-1}) Q_{n-1}$$

$$Q_{n+1} = \sum_{i=1}^n \alpha_i R_i \prod_{j=i+1}^n (1-\alpha_j) = Q_{n+1}$$

3) $Q_n \triangleq \frac{R_1 + \dots + R_n}{n-1}$ estimate of q^*

biased if $E[Q_n] \neq q^*$

② sample-average $Q_t(a) = \frac{\sum_{i=1}^{t-1} R_i \cdot \mathbb{1}_{A_i=a}}{\sum_{i=1}^{t-1} \mathbb{1}_{A_i=a}}$ as $t \rightarrow \infty$
 $N(A) \rightarrow \infty \rightarrow \sum_{i=1}^{t-1} \mathbb{1}_{A_i=a} \rightarrow \infty$

• law of large numbers, as $N(A) \rightarrow \infty$ then $Q_t(a) \rightarrow q^*(a)$
 it's unbiased if the amount actions taken (per each action) approaches infinity, others biased

Assume $0 < \alpha < 1$ $Q_{n+1} = Q_n + \alpha[R_n - Q_n]$

③ if $Q_1 = 0$ is Q_n for $n > 1$ biased? fraction R

$0 + \alpha[R_1 - 0] \Rightarrow Q_2 = \alpha R_1$

$Q_3 = \alpha R_1 + \alpha[R_2 - \alpha R_1] \Rightarrow \alpha R_1 + \alpha R_2 - \alpha^2 R_1$

$Q_3 = \alpha R_1(1-\alpha) + \alpha R_2$ $Q_4 = Q_3 + \alpha[R_3 - Q_3]$

$Q_4 = \alpha R_1(1-\alpha) + \alpha R_2 + \alpha[R_3 - \alpha R_1(1-\alpha) - \alpha R_2]$

$Q_4 = \alpha R_1(1-\alpha) + \alpha R_2 + \alpha R_3 - \alpha^2 R_1(1-\alpha) - \alpha^2 R_2$

$Q_4 = \alpha R_1(1-\alpha) - \alpha^2 R_1(1-\alpha) + \alpha R_2(1-\alpha) + \alpha R_3$

$\alpha R_1 - \alpha^2 R_1 - \alpha^2 R_1 + \alpha^3 R_1 + \alpha R_2 - \alpha^2 R_2 + \alpha R_3$

$Q_4 = \alpha R_1(\alpha^2 - 2\alpha + 1) + \alpha R_2(1-\alpha) + \alpha R_3$

previous rewards tend to 0 as $\alpha \rightarrow 1$ closer α to 1 closer R_3 is correct

the most heavily weighted term is the most recent term, which will be biased due to the undiminished α term as different

$$3) Q_n \triangleq \frac{R_1 + \dots + R_n}{n-1}$$

$$Q_{n+1} = Q_n + \alpha [R_n - Q_n]$$

② $\beta_n \triangleq \alpha / \bar{\alpha}_n$ to process n^{th} reward of particular action, where $\alpha > 0$ is const.
 $\bar{\alpha}_n$ is a trace if one that sticks at $\emptyset$:

$$\bar{\alpha}_n = \bar{\alpha}_{n-1} + \alpha(1 - \bar{\alpha}_{n-1}), \text{ for } n > 0 \text{ with } \bar{\alpha}_0 = \emptyset$$

$$Q_{n+1} = Q_n + \beta(R_n - Q_n)$$

$$= Q_n + \beta R_n - \beta Q_n \rightarrow Q_n(1 - \beta) + \beta R_n$$

$$Q_{n+1} = Q_n(1 - \frac{\alpha}{\bar{\alpha}_n}) + \frac{\alpha}{\bar{\alpha}_n} R_n \rightarrow \bar{\alpha}_n Q_{n+1} = \alpha R_n + \bar{\alpha}_n Q_n - \alpha Q_n$$

$$\bar{\alpha}_n Q_{n+1} = \alpha R_n - (\bar{\alpha}_n - \alpha) Q_n$$

$$\bar{\alpha}_n Q_{n+1} = \alpha R_n - [\bar{\alpha}_{n-1} + \alpha(1 - \bar{\alpha}_{n-1}) - \alpha] Q_n$$

$$= \bar{\alpha}_{n-1} Q_n + \alpha Q_n - \alpha \bar{\alpha}_{n-1} Q_n - \alpha Q_n$$

$$\bar{\alpha}_n Q_{n+1} = \alpha R_n - (1 - \alpha) \bar{\alpha}_{n-1} Q_n$$

$$Q_{n+1} = (1 - \alpha) Q_n + \sum_{i=1}^n \alpha(1 - \alpha)^{n-i} R_i$$

③

$$\bar{\alpha}_0 = \emptyset$$

$$\bar{\alpha}_1 = \emptyset + \alpha(1 - \emptyset)$$

$$\bar{\alpha}_1 = \alpha$$

$$\bar{\alpha}_2 = \alpha + \alpha(1 - \alpha)$$

$$= \alpha + \alpha - \alpha^2$$

$$= 2\alpha - \alpha^2$$

⋮

$$\bar{\alpha}_n Q_{n+1} = (1 - \alpha)^n \bar{\alpha}_0 Q_1 + \sum_{i=1}^n \alpha(1 - \alpha)^{n-i} R_i \rightarrow \bar{\alpha}_0 = \emptyset$$

$$\bar{\alpha}_n Q_{n+1} = \sum_{i=1}^n \alpha(1 - \alpha)^{n-i} R_i$$

the initial Q_1 value is canceled out by the $\bar{\alpha}_0$ value of $\emptyset$

$$Q_{n+1} = (1 - \alpha)^n Q_1 + \sum_{i=1}^n \alpha(1 - \alpha)^{n-i} R_i$$

$$\sum_{i=1}^n \alpha r^{n-i} = \frac{a}{1-r}$$

$$a = \alpha R_i$$

$$r = (1 - \alpha)$$

as $n \rightarrow \infty$
 $(1 - \alpha)^n$ approaches $\emptyset$

$$\frac{\alpha R_i}{1 - 1 + \alpha} \Rightarrow \frac{\alpha R_i}{\alpha} \Rightarrow R_i$$

geometric sum

sum of R_i terms, left other factors become much less valued, diminish.

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3) $\downarrow$
 ⑥ $Q_{n+1} = Q_n + \alpha[R_n - Q_n]$ $Q_2 = Q_1 + \alpha[R_1 - Q_1]$

$Q_3 = Q_2 + \alpha[R_2 - Q_2]$

$Q_3 = Q_1 + \alpha[R_1 - Q_1] + \alpha[R_2 - [Q_1 + \alpha[R_1 - Q_1]]]$

$\Rightarrow Q_1 + \alpha R_1 - \alpha Q_1 + \alpha R_2 - \alpha(Q_1 + \alpha[R_1 - Q_1])$

$\Rightarrow Q_1 + \alpha R_1 - \alpha Q_1 + \alpha R_2 - \alpha Q_1 + \alpha^2 R_1 + \alpha^2 Q_1$

$\alpha^2 R_1 + \alpha R_1 + \alpha R_2 + Q_1 - \alpha Q_1 - \alpha Q_1 + \alpha^2 Q_1$

$\Rightarrow Q_1(1 - 2\alpha + \alpha^2)$

$\Rightarrow \alpha R_1(\alpha + 1) + \alpha R_2 + Q_1(1 - 2\alpha + \alpha^2)$

$\cdot \alpha = 0 \rightarrow$ only Q_1 term remains

$\cdot$ as $n \rightarrow \infty$ Q_1 choice still has effect on all future Q estimates

the initial value of Q is still in the value of Q to stay in

the value of Q is still in the value of Q to stay in

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5. The different agents converge to different values based on the epsilon values. The initial rewards look better for the agent with the higher exploration value, but as time goes on and the Q values approach the correct values it will be making explorative actions when it no longer really needs to, lower the cumulative reward it could be receiving. The greedier option (that is not $\epsilon=0$) converges to $1 - \epsilon = 1 - 0.01 = 0.99$, and $\epsilon = 1 - 0.1 = 0.9$. The first one is 0.09 better.

6. The mean max q value is the theoretical max average value that can be received as reward for the agent, each trial is aiming to reach that value, maximize reward.

The cumulative rewards have not reached the predicted convergences, the more exploration heavy agent has risen faster than the greedier agent, but as more steps occurred the $\epsilon=0.01$ agent will eventually over take the $\epsilon=0.1$ agent.

7. The higher initial Q value results in much more exploration, since initially every option seems like it could be the best option and it greedily chooses the best option more often. But the Q values all need to be reduced from the optimistic value, which causes the agent to cycle through actions until they start to approach the correct Q values. UCB tries to pick actions it has not performed as much as others; it becomes more certain in its decisions and guesses on the action with the best upper bound. When it has little info, it makes a guess on which action has the highest upper bounds, but it has selected a worse option in terms of reward.

All graphs are included below, pdfs of them separate included as well



